

MIT THINK 2025-2026 Results

Overview

This year, we received a total of 1042 applications, the most ever in our history. Based on our evaluation of each applicant's proposal and fit for THINK, we selected 29 semifinalists to advance to the interview stage, and among those we selected five finalists. Finalist abstracts are available for view at the end of this document.

Additionally, due to the massive increase in the size of our applicant pool over the past few years, we felt that it is important to recognize the large number of very strong applications that we unfortunately don't have the capacity to advance. As such, we have introduced the new distinction of Commended to recognize the projects within the top 10% of our application pool who did not advance to semifinalist. This cycle, 75 projects were awarded this distinction.

Finalists

Moondust Magnet: A Bipolar Electrostatic System for Capturing Airborne Sub-micron Lunar Regolith in Habitat Airlocks
Jiaen Jiang | Thomas E. Randle High School, TX

Hybrid Lock: A Dynamically Stabilized Partial EMT State Underlying Aggressive ER-Low Breast Cancer and Population-Level Disparities
Audrey Guo | Torrey Pines High School, CA

Non-Invasive Structural Health Monitoring of a Hydraulic Prosthetic Knee Using Two-Channel Vibro-Acoustic Transmissibility
Hari Tadpatri | William Mason High School, OH

Enzymatically Crosslinked Lignin-Tyramine Bioadhesive Hydrogel for Real-Time Heavy Metal Sensing: A Novel Dual-Functional Material for Environmental Monitoring
Kireeti Kompella & Pranav Kokati | Interlake Senior High School, WA

Hybrid SiC–SiOC/Carbon Aerogels with ZrB_2 Skins for Advanced Spacecraft Thermal Protection
Amogh Vinaykumar | Flower Mound High School, TX

Semifinalists

Ronav G, CA
Kush D, NJ
Vishakha R, TX
Jasmine D, NY
Owen L, NY & Owen Z, NY
Audrey G, CA
Hari T, OH
Srihari K, IA
Advay S, WA
Jiyun L, VA & Vijay A, VA
Nathan Z, MO & Evan X, MO
Ivan H, CA
Swathi S, TX
Jiaen J, TX
Sehaj S, NY
Ananya G, MO
Kireeti K, WA & Pranav K, WA
Amogh V, TX
Sanjay S, WA
Anubhav M, TX
Kenechukwu (Kene) O, FL
Juhi B, CA
Kayla M, NY
Anna T, FL
Zixiang (Sophia) Z, NH
Illaria L, NY
Alexander D, PA & Mateen H, PA
Adia A, FL
Vriti M, TX & Niketa P, TX

Commended

Eric X, VA
Neil G, NJ
Samantha Y, CT
Arav R, CA & Frazer S, CA
Vaibhav S, NJ & Samaksh B, NJ
Matvei Y, MA & Grady M, MA
Niral P, CA & Shivendra S, CA
Rhea C, TX
Aashrith K, TX
Juan M, CA
David T, CA
Haripriya A, FL
Leen Z, AL & Zubayr R, AL
Leo W, CA & Jason W, CA
Venika K, CA
Aarnavi D, TX
Avelyn J, CA
Samika A, NJ & Ruhan A, NJ
Ameya T, CA
Tejas K, PA & Harivighnesh N, PA
Bhav J, FL
Ana B, TX
Aarav S, TX & Mrinal Y, TX
Arjun A, MD
Vidyuth P, CA
Saaketh B, IL & Arun M, IL
Reyna Y, CA & Valerie W, CA
Evelyn Z, MA & Sreshta T, MA
Mehul B, CA
Maanika K, CA
Micah A, VA
Frazer S, CA & Arav R, CA
Liam W, IL
Madhav K, NJ
Amber B, MN
Prisha S, TX
Alisha L, AZ
Ernest Q, FL
Prince N, TX
Eiden H, CA
William H, CA
Ayush V, MO
Chloe S, CA
Minjia H, VA & Yoonsub C, VA
Makaita M, GA
Shouryan M, CA
Aryan S, CT
Srikrishna D, NJ
Anvith A, FL & Abhi A, FL
Anirudh I, CA
Eric L, NJ & Gautham K, NJ
Sahana R, TX
Wanruo W, CA
Aiden K, PA & Rajan S, PA
Om V, CO
Skylar R, GA
Sharvin G, TX
Kelly L, CA
Luckison D, CA
Anya K, GA
Syeda Zainab F, CA
Eddie L, NY
Aiden C, TX
Kaveri D, MA & Anika A, MA
Ethan G, NY
Anya A, NC
Avyukth R, MO
Raya S, CA & Riya R, CA
Shambhvi D, NJ
Atharv K, IL
Kameron L, VA
Ruby Y, PA
Stefania L, CA
Suhana S, NJ
Samhita K, GA & Siruni P, GA

Finalist Abstracts

Moondust Magnet: A Bipolar Electrostatic System for Capturing Airborne Sub-micron Lunar Regolith in Habitat Airlocks

Jiaen Jiang | Thomas E. Randle High School, TX

Abstract: Lunar regolith poses a significant challenge to current and future space operations, as its properties have adverse effects on equipment and human health. While techniques for removing regolith from equipment have been developed, there is currently no system that is designed to extract airborne submicron regolith dust particles that may be introduced into the habitat through airlocks. The objective of this project is to devise and evaluate a novel bipolar electrostatic capture system designed to extract these airborne particles before it infiltrates the habitat. To achieve this, a custom high-voltage embedded bipolar electrode array will be prototyped and tested in an enclosed environment with various types of lunar dust stimulants from different areas of the moon to determine the efficiency of this technique in capturing airborne dust. The goal is for the system to be able to extract at least 80% of airborne lunar dust particles within an airlock, thereby reducing the influx of lunar dust into the habitat and improving crew safety.

Hybrid Lock: A Dynamically Stabilized Partial EMT State Underlying Aggressive ER-Low Breast Cancer and Population-Level Disparities

Audrey Guo | Torrey Pines High School, CA

Abstract: Black women with estrogen receptor–positive (ER+) breast cancer experience disproportionately aggressive disease that is not fully explained by clinical subtype or access to care. While Epithelial–Mesenchymal Transition (EMT) is a central driver of metastasis, EMT is increasingly recognized as a continuum rather than a binary process. Using RNA-sequencing data from TCGA breast tumors, I have constructed a Hybrid Lock index, a stable partial EMT state, defined by suppressed EMT completion (low ZEB1) alongside persistent epithelial junction expression (high EPCAM and CLDN3). I show that ER-low ER+ tumors cluster near a common point in gene-expression space, consistent with a shared partial EMT phenotype, and that the Hybrid Lock state is enriched in tumors from Black patients compared to White patients within biologically relevant strata.
This project proposes an integrated computational-experimental validation framework to determine whether this transcriptionally defined Hybrid Lock state corresponds to a dynamically stable regulatory attractor with functional consequences. I will use parametric ordinary differential equation (ODE) modeling to test for dynamic stability, then validate

predictions using TCGA proteomic data from the same patient cohort to confirm that transcriptional stability translates to sustained epithelial junction protein expression. This multi-level approach addresses a critical limitation of computational models: ensuring predicted stable states reflect biological function, not merely transient RNA patterns.

Non-Invasive Structural Health Monitoring of a Hydraulic Prosthetic Knee Using Two-Channel Vibro-Acoustic Transmissibility

Hari Tadpatri | William Mason High School, OH

Abstract: Hydraulic prosthetic knees contain structural interfaces such as pylons, adapters, and fasteners, plus damping elements that shape motion. Gradual changes in stiffness from interface loosening, or changes in damping from hydraulic drift, can be difficult to detect before they create instability or safety risk. This project proposes a low-cost bench-top diagnostic method that detects controlled early-stage changes using repeatable impact testing and two-channel measurement.

A gravity-driven swing rig produces impacts on a fixed durometer pad mounted to a steel stop plate. One piezo sensor measures the knee response on the rigid pylon or adapter, and a second piezo sensor measures a repeatability reference on the stop plate From accepted impacts, the analysis computes a frequency-domain transmissibility (response divided by reference) and magnitude-squared coherence to quantify consistency versus frequency, using coherence as a quality metric and the reference channel for per-impact acceptance gating.

Two phantom-powered high-impedance buffers condition the piezo signals and feed a two-input USB audio interface for recording. Datasets will be collected for baseline, controlled torque reduction at a single bench interface (simulated looseness), and controlled damping change using an external bench-safe damper if internal adjustment is not permitted. Goal: reliably distinguish baseline from controlled looseness/damping changes across days. Performance will be evaluated across different days using leave-one-session-out validation and reported with sensitivity, specificity, and a confusion matrix.

Enzymatically Crosslinked Lignin-Tyramine Bioadhesive Hydrogel for Real-Time Heavy Metal Sensing: A Novel Dual-Functional Material for Environmental Monitoring

Kireeti Kompella | Interlake Senior High School, WA

Abstract: Copper contamination remains a persistent challenge in water systems, particularly in industrial and low-resource settings where contamination events are intermittent and difficult to detect using conventional monitoring approaches. While laboratory techniques such as atomic absorption spectroscopy and inductively coupled plasma mass spectrometry offer high sensitivity, they require centralized facilities, trained personnel, and delayed analysis. Low-cost copper test kits support point-of-use testing but rely on manual sampling and cannot capture transient contamination events in flowing water systems. As water infrastructure becomes increasingly complex, there is a growing need for monitoring tools that can indicate when and where copper contamination occurs without continuous human oversight. This project proposes a passive, copper-specific hydrogel sensor designed to function as a deployable early-warning indicator. The goal is to develop a low-cost, disposable material that can be placed directly within pipes, drainage channels, or effluent streams and accumulate copper exposure over time, enabling localized screening and targeted follow-up testing. The sensing platform is based on an enzymatically crosslinked lignin–tyramine hydrogel incorporating a copper-selective colorimetric indicator bathocuproine disulfonate (BCS). Kraft lignin, an abundant industrial byproduct, serves as a redox-active matrix that localizes dissolved copper and facilitates its conversion to a detectable form. Horseradish peroxidase-mediated crosslinking produces a mechanically robust hydrogel capable of adhering to surfaces and remaining stable under flowing conditions. Upon cumulative exposure to copper, the hydrogel undergoes an irreversible visible color change, providing a simple and infrastructure-light method for in situ copper monitoring.

Hybrid SiC–SiOC/Carbon Aerogels with ZrB_2 Skins for Advanced Spacecraft Thermal Protection

Amogh Vinaykumar | Flower Mound High School, TX

Abstract: Reusable spacecraft require heat shields that are both extremely lightweight and capable of surviving repeated exposure to the intense stresses of atmospheric reentry. However, current solutions force a tradeoff: reinforced carbon–carbon is thermally durable but heavy, while ablatives like PICA are single-use, eroding under sustained heat flux. These limitations restrict payload capacity, compromise safety, and hinder the long-term viability

of affordable, sustainable spacecraft operations. This project aims to develop and evaluate a SiC-reinforced SiOC–carbon hybrid aerogel with a thin ZrB_2 emissive skin to overcome that tradeoff. The objective is a proof-of-concept TPS tile combining ultralow bulk density (target 0.15–0.35 $g \cdot cm^{-3}$), low thermal conductivity (target ≤ 0.08 $W \cdot m^{-1} \cdot K^{-1}$), and mechanical robustness sufficient for handling and mounting, while tolerating temperatures exceeding 1,500°C. Specifically, the project will follow a two-stage workflow. Stage 1 (home/basic lab): synthesis of polysiloxane preceramic resin co-gelled with dispersed SiC nanopowder and a kraft lignin carbon precursor, followed by aging, graded solvent exchange, and freeze-drying to produce a strong, monolithic structure. Stage 2 (supervised lab): controlled inert-gas pyrolysis (800–1000 °C) to convert the preceramic into a silicon oxycarbide backbone, then application of an ultrathin ZrB_2 slurry to improve oxidation resistance/stress control under high-heat conditions. Performance will be characterized through thermal-conductivity measurements, compressive-strength testing, and SEM microscopy to evaluate pore architecture and microcracking behavior. By combining polymer-derived ceramics with a high-temperature emissive coating, this work seeks to establish a low-cost pathway toward durable/reusable, lightweight heat-shield materials suitable for future spacecraft missions.